A Formal Proof of the Lieb–Robinson Bound and Exponential Clustering on Finite Lattices in Lean 4

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Abstract

The Lieb–Robinson bound states that in a quantum lattice system with local interactions, commutators of time-evolved local observables are exponentially small outside an effective light cone, with a velocity independent of the size of the system. We present a machine-checked proof of this bound for finite lattices in the Lean 4 proof assistant, on top of Mathlib. The setting is a C*-algebra with a subalgebra of observables for each region, subject only to the axiom that observables in disjoint regions commute, so no tensor products are needed. For an interaction of strength J, range R and coordination κ we prove $\|[\tau_t(A), B]\| \le 2e\,\|A\|\|B\|\,(n_X/\kappa)\,e^{v|t|-d(X,Y)/R}$ with $v = 2eJ\kappa$, where n_X is the number of interaction terms meeting the support of A. The proof removes the norm-preserving part of the dynamics with an explicit interaction-picture unitary, which avoids time-dependent propagators, and iterates the resulting integral inequality by induction on distance in the overlap graph of the interaction. We instantiate the abstract theorem for the concrete lattice model of the Physlib library. As the main application we prove the exponential clustering theorem: in a unique ground state with spectral gap γ, connected correlations of observables at distance d decay like $e^{-\mu d}$ with $\mu = \min(1/2, \gamma/(8eJ\kappa))/R$. The spectral filter in its proof is evaluated with Fubini's theorem and the Fourier transform of the Gaussian, without the error function or principal values. The development uses no axioms beyond Lean's standard three.

1 Introduction

Lieb and Robinson showed in 1972 that quantum spin systems with finite-range interactions have a finite group velocity [2]. Their bound became a central tool of quantum many-body physics only three decades later, when Hastings, Koma, Nachtergaele, Ogata, Sims and others used it to prove exponential clustering of correlations in gapped ground states [4, 1], to control the propagation of correlations [3], and, among many other results, to prove the higher-dimensional Lieb–Schultz–Mattis theorem and the stability of topological order. A survey is [5].

Despite this central role, we are not aware of a formalization of the Lieb–Robinson bound in any proof assistant. Before starting we searched the web, the Physlib library [6] and its open pull requests and issues. Physlib contains the kinematics of quantum lattice systems (nets of local algebras, the quasi-local algebra, interactions, finite-volume Hamiltonians and Gibbs states) but no time evolution and no locality estimate. This note fills that gap for finite lattices.

*AI system (model Claude Opus 5.5), developed by Anthropic. Claude wrote the Lean proofs and drafted this manuscript in collaboration with the first author.

Contributions.

- An abstract setting that uses only the commutation of observables in disjoint regions (§2).
- The basic integral inequality, proved with an explicit interaction-picture unitary instead of a time-dependent propagator (§3).
- Its iteration by induction on a recursive notion of distance in the overlap graph of the interaction, without matrices or explicit chains (§4).
- The bound in the familiar metric form $e^{v|t|-d/R}$ (§5), and its instantiation for Physlib's concrete lattice model (§6).
- The exponential clustering theorem for gapped ground states (§7).

The source is available in the `lieb-robinson` folder of `https://github.com/Crotalus/ai-formalized`; it builds against Physlib commit `3bfc7af` and Lean `v4.35.0`.

2 Setting

Let Λ be a finite set of sites and $\mathfrak{A}$ a C*-algebra. Physically $\mathfrak{A}$ is the algebra of all observables of the system, for instance $\bigotimes_{x\in\Lambda} M_{n_x}(\mathbb{C})$. The usual construction through tensor products is cumbersome in current Mathlib, and the Lieb–Robinson bound needs very little of it. A *local net* assigns a subalgebra $\mathfrak{A}_X \subseteq \mathfrak{A}$ to every region $X \subseteq \Lambda$, and we assume only that observables in disjoint regions commute:

```
structure LocalNet (Λ 𝔄 : Type*) [CStarAlgebra 𝔄] where
  alg : Finset Λ → Subalgebra ℂ 𝔄
  commute_of_disjoint : ∀ {X Y : Finset Λ}, Disjoint X Y →
    ∀ a ∈ alg X, ∀ b ∈ alg Y, Commute a b

structure Interaction (N : LocalNet Λ 𝔄) where
  Φ : Finset Λ → 𝔄
  mem_alg : ∀ Z, Φ Z ∈ N.alg Z
  isSelfAdjoint : ∀ Z, IsSelfAdjoint (Φ Z)
```

We do not even assume that $\mathfrak{A}_X \subseteq \mathfrak{A}_Y$ for $X \subseteq Y$: the proof never uses it. An interaction assigns a self-adjoint term $\Phi(Z) \in \mathfrak{A}_Z$ to each region. Its Hamiltonian is $H = \sum_Z \Phi(Z)$ and the Heisenberg dynamics is $\tau_t(A) = e^{itH} A e^{-itH}$, defined with Mathlib's exponential in Banach algebras. For self-adjoint K the exponential e^{itK} is unitary, so τ_t is a $*$-automorphism that preserves norms. The commutator is $[a, b] = ab - ba$.

The constants of the bound are:

- the *strength* J, with $\|\Phi(Z)\| \le J$ for all Z;
- the *coordination* κ: at most κ regions Z' with $\Phi(Z') \neq 0$ meet any region Z with $\Phi(Z) \neq 0$ (Lean: `(I.neighbours Z).card` $\le \kappa$);
- for the metric form, the *range* R: every non-zero term has diameter at most R.

3 The integral inequality

Lemma 1. *For $A \in \mathfrak{A}_X$, any $B \in \mathfrak{A}$ and $t \geq 0$,*

$$\|[\tau_t(A), B]\| \leq \|[A, B]\| + 2\|A\| \sum_{Z \cap X \neq \emptyset} \int_0^t \|[\tau_s(\Phi(Z)), B]\|\, ds.$$

Proof. Split $H = H_1 + H_2$, where H_2 collects the terms whose region meets X. Every term of H_1 lives in a region disjoint from X, so H_1 commutes with A and with e^{isH_1}. Put $U(t) = e^{itH}e^{-itH_1}$ and $V(t) = e^{itH_1}e^{-itH}$, so that $UV = 1$ and $\tau_t(A) = U(t)\,A\,V(t)$. Then

$$[\tau_t(A), B] = U(t)\,[A, W(t)]\,V(t), \qquad W(t) = V(t)\,B\,U(t),$$

and since $\|U\|, \|V\| \leq 1$ we get $\|[\tau_t(A), B]\| \leq \|[A, B]\| + 2\|A\|\,\|W(t) - W(0)\|$. A direct computation gives

$$W'(s) = e^{isH_1}\,[\tau_{-s}(B),\, iH_2]\,e^{-isH_1},$$

whose norm is at most $\|[H_2, \tau_{-s}(B)]\| = \|\tau_{-s}([\tau_s(H_2), B])\| \leq \sum_{Z \cap X \neq \emptyset} \|[\tau_s(\Phi(Z)), B]\|$. Integrating this bound on W' gives the claim. □

The textbook proof [4] obtains the same inequality from the differential equation $g'(t) = i[\tau_t(H_2), g(t)] + \dots$ for $g(t) = [\tau_t(A), B]$, using that the solution operator of $g' = i[K(t), g]$ is isometric. Formalizing this requires the time-dependent propagator of that equation. The interaction-picture unitary U achieves the same effect using only exponentials of fixed operators. A Grönwall argument applied directly to g would avoid both but gives a weaker inequality, in which the supports of later terms grow with each step; iterating it gives only a short-time bound.

In Lean the derivative of W is computed with `HasDerivAt.mul` for the noncommutative product of four factors and checked against the claimed form by `noncomm_ring`. The integrated bound is Mathlib's comparison theorem `image_norm_le_of_norm_deriv_right_le_deriv_boundary`, with the bounding function $t \mapsto \int_0^t g$ differentiated by the fundamental theorem of calculus.

```
theorem norm_commutator_evolution_le {X : Finset Λ} {A : 𝔄} (hA : A ∈ N.alg X) (B : 𝔄)
    {t : ℝ} (ht : 0 ≤ t) :
    ‖commutator (I.evolution t A) B‖ ≤ ‖commutator A B‖ +
      2 * ‖A‖ * ∑ Z ∈ touching X, ∫ s in (0 : ℝ)..t,
        ‖commutator (I.evolution s (I.Φ Z)) B‖
```

4 Iteration

Fix $B \in \mathfrak{A}_Y$ and write $f_Z(t) = \|[\tau_t(\Phi(Z)), B]\|$. Lemma 1 with $A = \Phi(Z)$ gives $f_Z(t) \leq f_Z(0) + 2\|\Phi(Z)\| \sum_{Z' \cap Z \neq \emptyset} \int_0^t f_{Z'}$, and $f_Z(0) = 0$ when Z is disjoint from Y. Rather than expanding this into a sum over chains, we measure distance to Y in the overlap graph by a recursive predicate:

```
def Far (Y : Finset Λ) : ℕ → Finset Λ → Prop
  | 0, _ => True
  | k + 1, Z => I.Φ Z = 0  (Disjoint Z Y  ∀ Z', (Z' ∩ Z).Nonempty → Far Y k Z')
```

$\mathrm{Far}\,Y\,k\,Z$ says that every chain of overlapping regions with non-zero terms from Z to a region meeting Y has at least k steps. A single induction on k then proves:

Lemma 2. *If $\mathrm{Far}\,Y\,k\,Z$ and $t \geq 0$, then $f_Z(t) \leq 2\|\Phi(Z)\|\|B\|\,(2J\kappa t)^k/k!$.*

The base case is $\|[a,b]\| \le 2\|a\|\|b\|$ together with $\|\tau_t(a)\| = \|a\|$. In the step, the terms with $\Phi(Z') = 0$ vanish, at most κ terms remain, and $\int_0^t (2J\kappa s)^k/k!\, ds = (2J\kappa)^k t^{k+1}/(k+1)!$. One more application of Lemma 1 gives the bound for a general A:

Theorem 3. *Let $A \in \mathfrak{A}_X$ and $B \in \mathfrak{A}_Y$ with $X \cap Y = \emptyset$, and suppose every region meeting X satisfies* $\operatorname{Far} Y\, n$. *Then for $t \ge 0$*

$$\|[\tau_t(A), B]\| \le 2\|A\|\|B\|\, n_X\, 2J\, (2J\kappa)^n\, \frac{t^{n+1}}{(n+1)!} \le 2\|A\|\|B\|\, \frac{n_X}{\kappa}\, e^{\,2eJ\kappa t - (n+1)},$$

where n_X is the number of non-zero terms meeting X (the second form needs $\kappa > 0$).

The exponential form uses $x^m/m! \le e^{ex-m}$, which follows from $(ex)^m/m! \le e^{ex}$. No infinite series is needed: the polynomial bound is already exact at finite order.

5 The metric form

Let Λ carry a (pseudo)metric and let every non-zero term have diameter at most $R > 0$. A chain of $j+1$ overlapping terms spans a distance of at most $(j+1)R$, so a region all of whose points are further than kR from Y is $\operatorname{Far} Y\, k$ (lemma `far_of_dist`). Choosing $n = \lceil d/R \rceil - 2$ (truncated at 0) yields:

Theorem 4 (Lieb–Robinson bound)**.** *If every point of X is at distance at least $d > 0$ from every point of Y, then for $A \in \mathfrak{A}_X$, $B \in \mathfrak{A}_Y$ and all real t,*

$$\|[\tau_t(A), B]\| \le 2e\, \|A\|\|B\|\, \frac{n_X}{\kappa}\, e^{\,v|t| - d/R}, \qquad v = 2eJ\kappa.$$

```
theorem lieb_robinson_abs {X Y : Finset Λ} {A B : 𝔄} (hA : A ∈ N.alg X) (hB : B ∈ N.alg Y)
    {J R d : ℝ} {κ : ℕ} (hκ0 : 0 < κ) (hJ0 : 0 ≤ J) (hJ : ∀ Z, ‖I.Φ Z‖ ≤ J)
    (hκ : ∀ Z, (I.neighbours Z).card ≤ κ) (hR0 : 0 < R)
    (hR : ∀ Z, I.Φ Z ≠ 0 → ∀ x ∈ Z, ∀ y ∈ Z, dist x y ≤ R)
    (hd0 : 0 < d) (hd : ∀ x ∈ X, ∀ y ∈ Y, d ≤ dist x y) (t : ℝ) :
    ‖commutator (I.evolution t A) B‖ ≤
      2 * Real.exp 1 * ‖A‖ * ‖B‖ * ((I.neighbours X).card / κ) *
        Real.exp (2 * Real.exp 1 * J * κ * |t| - d / R)
```

Negative times reduce to positive ones: the interaction $-\Phi$ satisfies the same hypotheses and generates τ_{-t}. None of the constants J, R, κ, n_X depends on $|\Lambda|$.

6 The concrete lattice model

Physlib models a lattice system with a finite configuration space Ω_s at each site s of a possibly infinite set S. The operators of a finite region $X \subseteq S$ are the complex matrices indexed by configurations of X, written `Alg X`, and a larger region $Y \supseteq X$ contains them through $M \mapsto M \otimes 1$ (`inclMat`). An interaction assigns a Hermitian operator to each finite region, and the Hamiltonian of a finite box Λ_0 is the sum of the lifts of the terms inside it. To apply the abstract theorem we fix a finite box Λ_0 and take:

- sites: the points of Λ_0, a finite type;

- algebra: `Alg  Λ₀`, a matrix algebra, with Mathlib's C*-norm (the operator norm for the ℓ^2 norm on configurations, `Matrix.Norms.L2Operator`);
- local algebras: the images of the lifts $M \mapsto M \otimes 1$ from each subregion.

The only property to verify is that lifts from disjoint regions commute. We prove this by computing both products $(M \otimes 1)(1 \otimes N)$ and $(1 \otimes N)(M \otimes 1)$ entrywise: each has a single non-zero term in the matrix product, at the configuration obtained by gluing.

```
lemma commute_inclMat_of_disjoint (hX : X ≤ Λ) (hY : Y ≤ Λ) (hd : Disjoint X Y)
    (M : Alg  X) (N : Alg  Y) : Commute (inclMat hX M) (inclMat hY N)
```

As a check that the two settings fit together, the Hamiltonian of the abstract interaction built from a Physlib interaction equals Physlib's Hamiltonian of the box (`hamiltonian_toBox`), after reindexing the sum over subsets of the box. The resulting corollary states the Lieb–Robinson bound directly for Physlib's operators $A \in$ `Alg  X` and $B \in$ `Alg  Y` with $X, Y \subseteq \Lambda_0$, the Physlib Hamiltonian, and strength and range hypotheses on the Physlib interaction. Since the lift is contractive, the norms of A and B may be taken in their own regions. The coordination constant is stated for the box interaction; formulating it on the infinite set of regions of S is a matter of bookkeeping.

7 Exponential clustering

The most important consequence of the Lieb–Robinson bound is that a spectral gap forces correlations to decay exponentially [4, 1]. We take $\mathfrak{A} = \mathcal{B}(\mathcal{H})$ for a finite-dimensional Hilbert space $\mathcal{H}$, which Mathlib equips with its C*-algebra structure.

Definition 5. A *gapped ground state* of a self-adjoint H with gap $\gamma > 0$ is a unit vector Ω with $H\Omega = E_0\Omega$ such that $\langle \varphi, H\varphi \rangle \geq (E_0 + \gamma)\|\varphi\|^2$ for all $\varphi \perp \Omega$. The connected correlation is $\langle AB \rangle - \langle A \rangle \langle B \rangle$, with $\langle X \rangle = \langle \Omega, X\Omega \rangle$.

The proof has three independent parts.

Spectral expansion. Expanding in an orthonormal eigenbasis φ_j of H, with energies $E_0 + \omega_j$, the function $F(t) = \langle [\tau_t(A), B] \rangle$ becomes

$$F(t) = F(0) + \sum_j c_j \, (e^{-i\omega_j t} - 1) - \sum_j d_j \, (e^{i\omega_j t} - 1),$$

with $\sum_j c_j = \langle AB \rangle - \langle A \rangle \langle B \rangle$, $\sum_j |c_j|, \sum_j |d_j| \leq \|A\|\|B\|$ (Cauchy–Schwarz and Parseval), and only frequencies $\omega_j \geq \gamma$ contributing. The last point uses the gap: an eigenvector with energy below $E_0 + \gamma$ lies in the span of Ω, so it does not contribute to the connected parts. To state the expansion with a single frequency list we replace each ω_j by $\max(\omega_j, \gamma)$, which changes only terms with zero coefficient. In Lean this needs the action of the exponential on eigenvectors, $e^K v = e^\mu v$ for $Kv = \mu v$, which we prove from the exponential series.

A Gaussian filter. The positive-frequency part $\sum_j c_j$ is extracted from F by integrating against $e^{-\alpha t^2}/(it)$. Since $F(0) = \langle [A, B] \rangle = 0$ for disjoint supports, it suffices to know

$$\int_{\mathbb{R}} e^{-\alpha t^2} \, \frac{e^{-i\omega t} - 1}{it} \, dt = -\sqrt{\pi/\alpha} \int_0^\omega e^{-\sigma^2/4\alpha} \, d\sigma.$$

We prove this by writing $(e^{-i\omega t}-1)/(it) = -\int_0^\omega e^{-i\sigma t}\,d\sigma$, exchanging the integrals and using Mathlib's Fourier transform of the Gaussian. The right-hand side is odd in ω and within $(2\sqrt{\pi\alpha}/\gamma)\,e^{-\gamma^2/4\alpha}$ of $-\pi$ for $\omega \geq \gamma$. This avoids the error function and principal-value integrals, neither of which Mathlib provides. Combining with the expansion gives $\int e^{-\alpha t^2}F(t)/(it)\,dt = -2\pi\sum_j c_j + (\text{error})$.

Time-domain bound. The same integral is bounded by $\int e^{-\alpha t^2}|F(t)|/|t|\,dt$. For $|t| \leq T$ the Lieb–Robinson bound in polynomial form gives $|F(t)| \leq K|t|^{n+1}$; for $|t| > T$ we use $|F| \leq 2\|A\|\|B\|$ and the Gaussian tail. Choosing $T = (n+1)/(4eJ\kappa)$ and $\alpha = \gamma/(2T)$ balances the three errors and yields:

Theorem 6 (Exponential clustering)**.** *Let Ω be a gapped ground state of $H = \sum_Z \Phi(Z)$ with gap γ, let $A \in \mathfrak{A}_X$ and $B \in \mathfrak{A}_Y$ with $X \cap Y = \emptyset$, and suppose every region meeting X satisfies* $\operatorname{Far} Y\, n$*. If $J, \kappa > 0$, then*

$$|\langle AB\rangle - \langle A\rangle\langle B\rangle| \leq \frac{2}{\pi}\|A\|\|B\|\Big(\frac{n_X}{\kappa}e^{-(n+1)/2} + \Big(\frac{4eJ\kappa}{\gamma} + \sqrt{\frac{2\pi eJ\kappa}{\gamma}}\Big)e^{-\gamma(n+1)/(8eJ\kappa)}\Big).$$

On a metric lattice with range R and $d(X,Y) \geq d$, the right-hand side becomes exponentially small in d at rate $\min(1/2, \gamma/(8eJ\kappa))/R$.

```
theorem exponential_clustering {γ : ℝ} (hγ : 0 < γ)
    (G : GappedGroundState I.hamiltonian γ) {X Y : Finset Λ} {A B : ℋ →L[ℂ] ℋ}
    (hA : A ∈ N.alg X) (hB : B ∈ N.alg Y) (hXY : Disjoint X Y) {J : ℝ} {κ : ℕ}
    (hκ0 : 0 < κ) (hJ0 : 0 < J) (hJ : ∀ Z, ‖I.Φ Z‖ ≤ J)
    (hκ : ∀ Z, (I.neighbours Z).card ≤ κ) {n : ℕ}
    (hn : ∀ Z, (Z ∩ X).Nonempty → I.Far Y n Z) :
    ‖G.corr A B‖ ≤ 2 / Real.pi * ‖A‖ * ‖B‖ *
      ((I.neighbours X).card / κ * Real.exp (-(n + 1) / 2) +
        (4 * Real.exp 1 * J * κ / γ + Real.sqrt (2 * Real.pi * Real.exp 1 * J * κ / γ)) *
          Real.exp (-γ * (n + 1) / (8 * Real.exp 1 * J * κ)))
```

The rate $\gamma/(8eJ\kappa) \sim \gamma/v$ is of the form γ/v up to constants, as in the rates obtained by Nachtergaele and Sims and by Hastings and Koma [4, 1]; we did not optimize constants. We assume a unique ground state. The degenerate case, where correlations are taken in a ground state projection, is not treated.

8 Discussion

Size and checking. The development consists of 18 definitions and 85 lemmas in about 2200 lines, including documentation, in nine files: the setting (§2), the integral inequality (§3), the iteration (§4), the metric form (§5), the lattice model (§6), and four files for exponential clustering (§7). `#print axioms` on the main results reports only `propext`, `Classical.choice` and `Quot.sound`. We have not run Physlib's contribution linters on these files.

Design choices. Three decisions kept the Lieb–Robinson part short. First, the axiom that disjoint regions commute replaces tensor products and isotony of the net; the bound uses nothing else about locality. Second, the interaction-picture unitary replaces the time-dependent propagator of the textbook proof. Third, the recursive predicate Far replaces the sum over chains: each

induction step needs only one application of the integral inequality and a count of neighbours. The proof was developed by first fixing all statements and checking them with `sorry`, after which the files were proved in parallel by separate instances of Claude against the frozen interface, and the lattice model was added in the same way. Every claim was then re-verified by a full build.

Constants. Our velocity $v = 2eJ\kappa$ has the same structure as in the literature [4, 5], where v is proportional to the interaction strength and a local summability constant. The factor e comes from the elementary estimate $x^m/m! \le e^{ex-m}$. The prefactor depends on the number n_X of terms meeting X, as is standard. We did not optimize constants.

Limitations and future work. The result is for finite lattices. The bounds are uniform in the volume, which is what the infinite-volume theory needs, but we do not construct the infinite-volume dynamics. Interactions are of finite range: exponentially decaying interactions would require the summability ("F-function") formalism of [3]. Exponential clustering is proved for a unique gapped ground state in finite dimension; the degenerate case and the infinite-volume formulation, where the gap is a property of a thermodynamic limit, are not covered. Further applications such as area laws or the Lieb–Schultz–Mattis theorem would require substantially more infrastructure.

Author contributions. N.L. chose the problem, directed the work and reviewed the results. Claude, an AI system developed by Anthropic, surveyed the literature and libraries, designed the statements and the proof architecture, wrote the Lean proofs (in part through parallel sub-agents) and drafted the manuscript. All statements were checked by the Lean kernel. As an AI system cannot take responsibility for a publication, N.L. is accountable for its content.

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